Lake Water Levels and Local Economic Activity: IV Evidence from Africa and South America

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ABSTRACT

Water availability is a key channel through which climate variability affects economic activity in low- and middle-income countries. This paper estimates the causal effect of lake water levels on nearby economic activity using monthly satellite data for 22 undammed lakes across Africa and South America. To address endogeneity, I instrument lake levels with upstream rainfall, controlling for local rainfall and fixed effects. The results show that short-run, rainfall-driven increases in lake levels significantly raise nearby night-time lights, with stronger effects around larger and more economically active lakes, supported by vegetation and conflict evidence. Overall, they highlight the economic importance of surface water variability for local economic activity and adaptation policy, and the short-run impacts of hydrological shocks.

Section 1. Introduction

Water availability is a key channel through which the climate affects livelihoods, where temperature and precipitation shocks are shown to reduce income and productivity in poorer countries (Dell, Jones, and Olken 2014). Lakes play a central role, supporting irrigated agriculture, fisheries, transportation and energy generation (Barbier et al. 2011; World Bank 2009). More recently, due to climatic factors, there have been large losses in global surface water, where more vulnerable regions face stronger effects (Woolway et al. 2020; Pekel et al. 2016).

Existing research focuses on one region, lake or historical time period, emphasising long-run shifts rather than contemporaneous economic activity (Jedwab et al. 2023; Rogers, Saginor, and Jithitikulchai 2014; Li et al. 2019). Due to this, there remains a gap in the literature on how lake levels affect high-frequency, short-run local economic activity spanning multiple regions. Therefore, this paper asks 'What is the causal effect of a change in lake water levels on nearby economic activity?' where I use satellite-measured monthly lake levels (Copernicus Climate Change Service, Lake Water Levels from 1992 to Present Derived from Satellite Observation) and monthly nighttime lights (Elvidge et al. 2017) as a proxy for economic activity across 22 lakes in Africa and South America, where formal economic data is sparse.

However, lake levels are endogenous and respond to economic activity through irrigation demand and water extraction, leading to a reverse causality problem biasing OLS estimates. To mitigate this, this paper uses monthly upstream rainfall

(Funk et al. 2015), which affects local lake levels but remains plausibly exogenous to local economic conditions (Miguel, Satyanath, and Sergenti 2004). Overall, this strategy isolates supply-driven variation in lake levels, allowing for causal high-frequency effects on economic activity in regions and rural communities that remain critically understudied.

The results show that higher lake levels increase local economic activity. IV estimates indicate that a one standard deviation increase in within-lake water levels leads to a statistically significant rise in nearby nighttime lights. Findings are robust across alternative specifications, spatial definitions, and mechanism tests, showing that higher lake levels increase vegetation. Conflict rises with sustained long-run lake expansion but falls following short-run inflows.

Section 2. Literature review and contribution

A domain of literature in development economics focuses on climate variability, water availability and the use of rainfall-based identification for economic outcomes. Dell, Jones, and Olken (2012) find an increase in temperature reduces per capita income, but only in poorer countries. Using rainfall shocks, Barrios, Bertinelli, and Strobl (2010) show a significant impact of rainfall decline on African countries, with declining rainfall explaining 15-40% of the African income gap relative to other developing countries. Miguel, Satyanath, and Sergenti (2004) use rainfall variation as an instrument for economic growth in sub-Saharan Africa, finding a significantly negative relationship between instrumented GDP-growth and civil conflict. Together, this shows climatic shocks are significant to economic growth, developing countries are more vulnerable and reinforces the use of exogenous rainfall-based variation for identifying causal effects.

Another domain of literature shows the significance of lakes on local communities and economies, especially in developing nations. Jedwab et al. (2023) find that the shrinking of Lake Chad led to substantially lower long term growth, with locations closer to the lake growing 45% slower in Chad. This, combined with their spatial model, implies a 6% aggregate welfare loss. Carey, Dickes, and Crouch (2016) find significant positive effects on local employment, output and disposable income from a 1ft increase in lake level, with gains varying by county proximity to the lake. Wider work also exists but is largely associative rather than causal (Li et al. 2019; Yu et al. 2023).

A growing body of literature also validates nighttime lights (NTL) as a proxy for economic activity. Henderson, Storeygard, and Weil (2012) show that changes in luminosity track changes in GDP over time. More recent work shows that NTL remains positively correlated with economic activity in rural areas in Colombia (Pérez-Sindín, Chen, and Prishchepov 2021). Together, this supports the use of NTL as a proxy for economic activity in rural areas where data remains sparse.

This paper extends the existing lake-based studies (which are focused on a single lake or region) by looking at 22 lakes across Africa and South America, allowing for inference across heterogeneous contexts. Additionally, much of the previous literature focuses on longer-term welfare, with short-term work being mainly associative. This paper estimates monthly effects of lake level variation on

local economic activity proxied by NTL. Finally, this paper addresses endogeneity by exploiting upstream rainfall as an instrument for local lake levels, isolating exogenous variation and controlling for local rainfall. This builds on rainfall-IV literature, by applying it to lake-levels for causal estimates.

Section 3. Data

I combine satellite-derived lake levels, nighttime lights, gridded rainfall and settlement maps to build a panel (2012-2025) linking monthly lake level variation to nearby nighttime lights. For robustness, I test lake level variation on local agriculture by using satellite-derived vegetation indices and consider conflict as an alternative outcome using geocoded conflict data. The unit of observation is lake-month with outcomes measured in a 10km ring around each lake's shoreline. This ring captures communities directly exposed to lake-based livelihoods.

Monthly lake levels are satellite-derived via the Copernicus Climate Data Store (Copernicus Climate Change Service, Lake Water Levels from 1992 to Present Derived from Satellite Observation), spanning 1992-present (however, time periods differ by lake). Lake level (L_{it}) is measured in metres, but for comparability, I Z-standardise:

$$ZL_{it} = \frac{L_{it} - \bar{L}_i}{\sigma_{L_i}}$$

This then forms the main endogenous regressor, where coefficients can be interpreted as the effect of one-within-lake standard-deviation change.

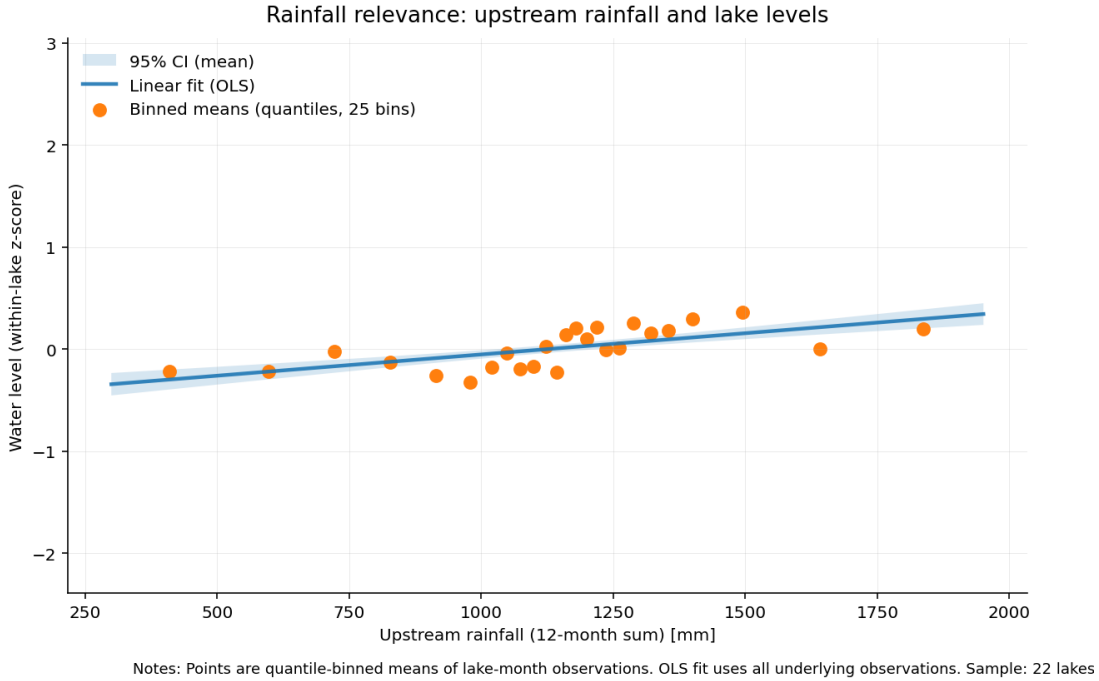
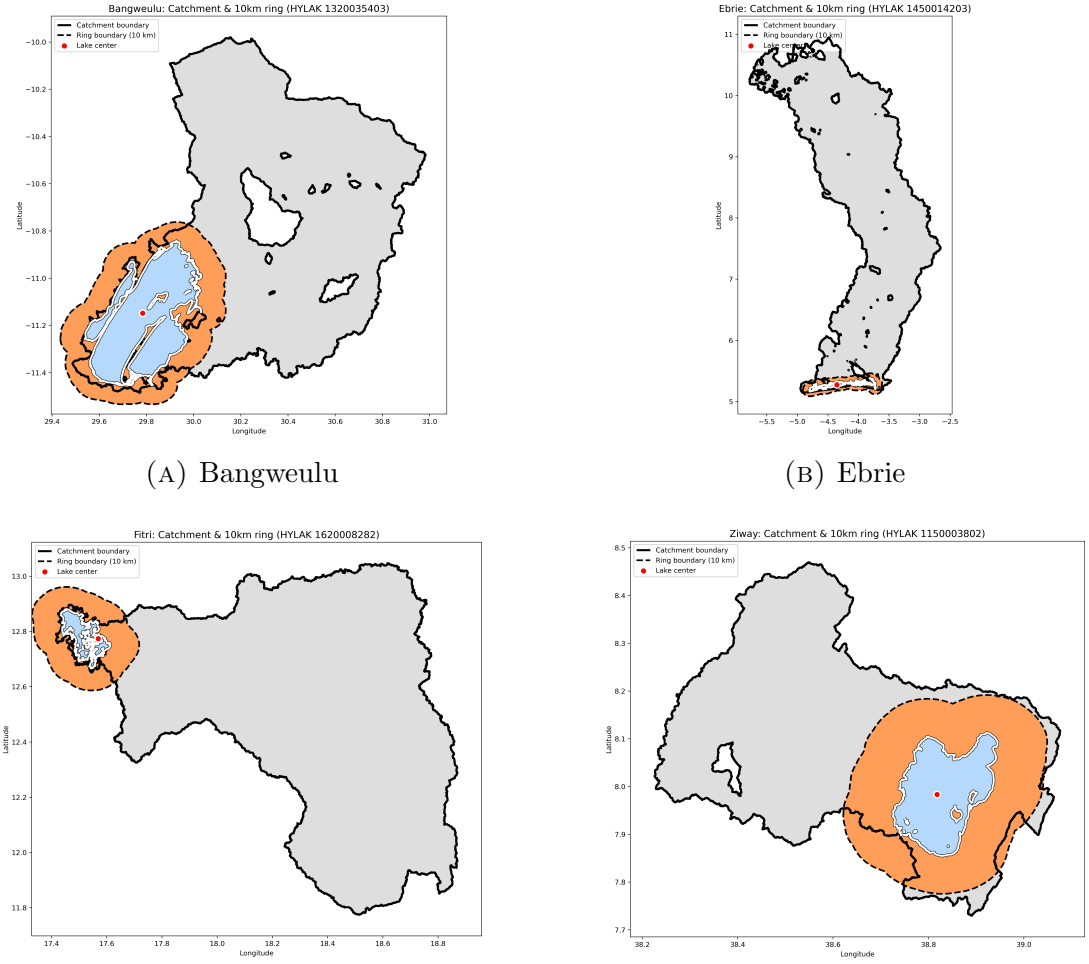


Figure 1. The shaded blue band is the 95% confidence interval for the mean trend. Lake levels show significant within-lake variation and are influenced by inflows, making it well-suited to an IV-strategy.

Rainfall is derived from CHIRPS (Funk et al. 2015), a global gridded precipitation dataset derived from satellites and rain gauges, available monthly. Let $Rain_{pt}$ denote rainfall in pixel p at time t. Then upstream rainfall is $\text{Rain}_{it} = \sum_{p \in c_i} \text{Rain}_{pt}$ where c_i denotes upstream-catchment areas for lake i. I similarly aggregate rainfall within the 10km ring as a local control for direct weather effects. With fine spatial resolution (0.05°) and high frequency, a distinct local and upstream rainfall polygon can be constructed for each lake.

For each lake, the upstream catchment polygon is from the Lake-TopoCat hydrological network (Sikder et al. 2023), capturing the drainage basin feeding inflows. Upstream catchments are then defined as catchments outside the 10km outcome-ring.



(A) Bangweulu

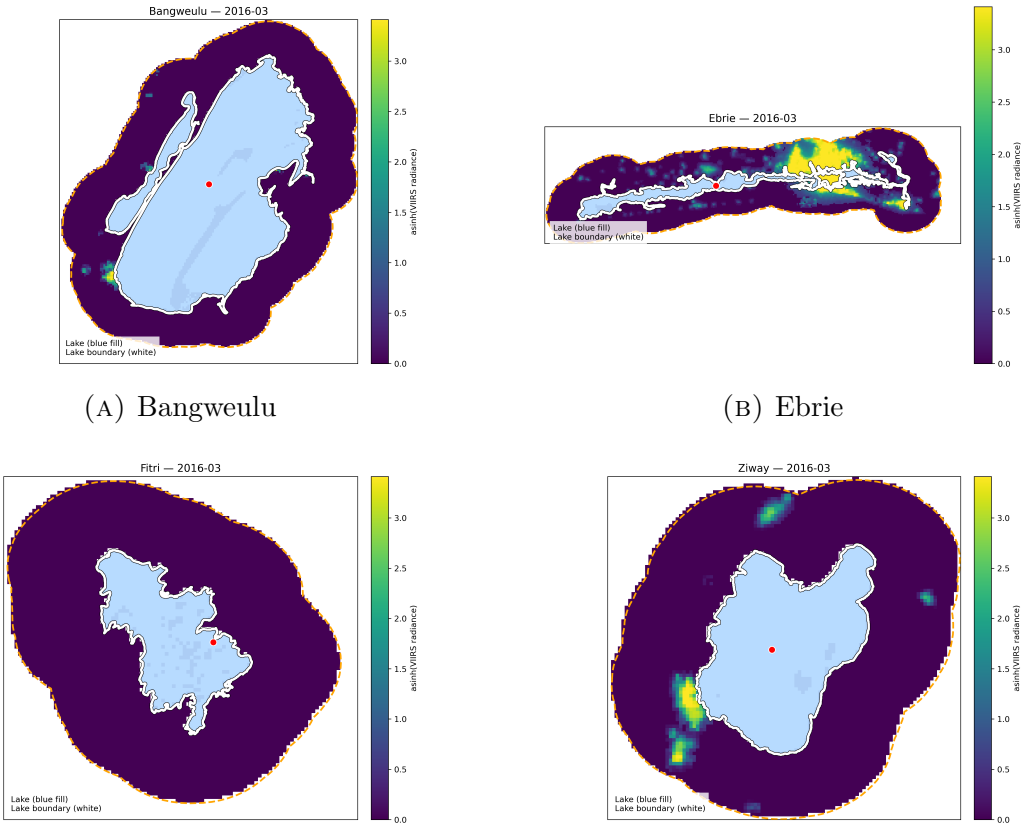
(B) Ebrie

(C) Fitri

(D) Ziway

FIGURE 2. Lake outcome rings (orange) and upstream catchments (grey). Note upstream catchments are catchments not within outcome rings.

Economic activity is proxied via monthly VIIRS night-time lights radiance (Elvidge et al. 2017). For each lake, I construct a 10km ring in which dependent communities lie. To classify pixels in this 10km ring into settled areas vs uninhabited land, I use the Global Human Settlement Layer (GHSL) (Pesaresi et al. 2024). This ensures the analysis focuses on economically meaningful pixels. This creates the primary outcome NL_{it}.

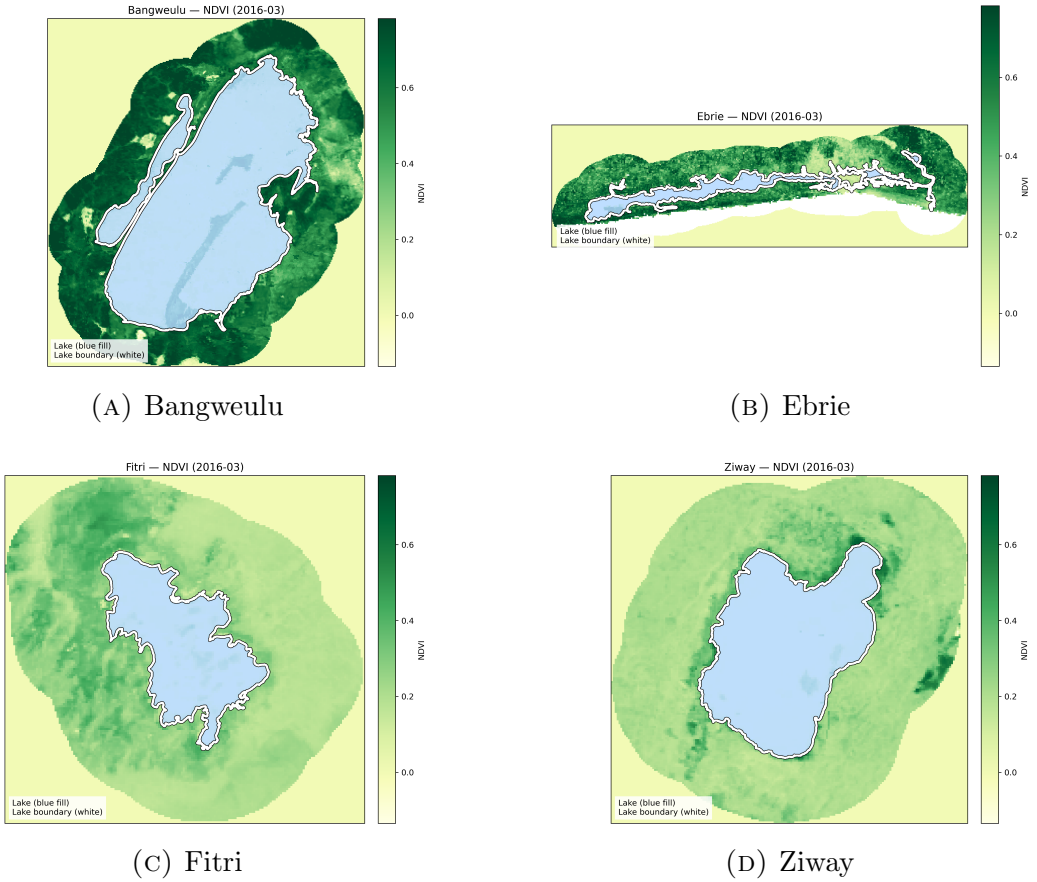


(A) Bangweulu (B) Ebrie

(C) Fitri (D) Ziway

FIGURE 3. Night-time lights around selected lakes (March 2016)

To consider a mechanism, I use the vegetation index MODIS NDVI (Didan, MOD13A3 MODIS/Terra Vegetation Indices Monthly L3 Global 1km SIN Grid V006). NDVI captures vegetation health that may respond to water availability. Further, I also consider an alternative outcome of geocoded conflict data (Raleigh, Kishi, and Linke 2023). The secondary outcome is built as the $log(1+conflict_{it})$, where $conflict_{it}$ represents the sum of conflict-events within the 10km ring at lake i time t.



(A) Bangweulu (B) Ebrie

(C) Fitri (D) Ziway

FIGURE 4. Vegetation (NDVI) around selected lakes (March 2016)

Finally, I only consider undammed lakes where no artificial water regulation is present. This ensures that lake levels are not driven by endogenous management choices. The final analysis contains 22 undammed lakes in Africa and South America with 2184 observation lake-months.

TABLE 1. Summary statistics

Variable	N	Mean	SD	p10	p50	p90	Min	Max
Nightlights mean (10km)	2184	1.179	3.259	0.000	0.202	2.277	-0.035	22.059
Lake level	2184	756.104	566.420	62.366	621.063	1,639.124	0.100	1,888.300
Upstream rainfall (12m, lag 1)	2024	1,118.658	361.367	610.323	1,139.159	1,579.756	242.131	2,294.347
Ring rainfall (12m, lag 1)	2024	1,033.665	384.386	527.169	1,037.182	1,577.131	181.830	2,372.157
NDVI mean (10km)	2184	0.424	0.166	0.200	0.407	0.665	-0.020	0.779
ACLED events (10km)	2184	0.116	0.551	0.000	0.000	0.000	0.000	10.000
ACLED fatalities (10km)	2184	0.094	1.217	0.000	0.000	0.000	0.000	46.000
ACLED $\log(1+\text{events})$ (10km)	2184	0.063	0.250	0.000	0.000	0.000	0.000	2.398

Notes: Observations are lake–months (2,184) across 22 undammed lakes. NDVI is a MODIS vegetation index averaged over a 10 km ring around each lake. Conflict variables are from ACLED and measured within 10 km of the lake. VIIRS night lights are often zero-coded in rural pixels (so small activity changes may be missed, biasing effects toward zero) and can exhibit blooming (so nearby bright sources may inflate measured light).

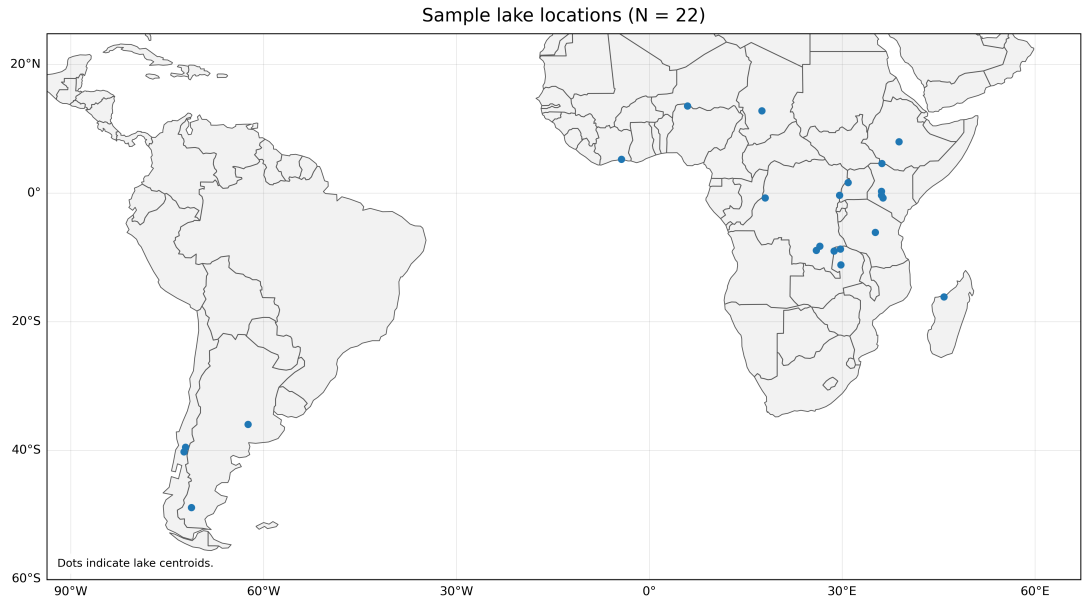


FIGURE 5. Map of Sample Lakes (Istifanos, Albert, Edouard, Mweru, Chishi, Bangweulu, Hinojo, Calafquén, Ranco, Cardiel, Gagere, Fitri, Ziway, Langano, Ebrie, George, Tumba, Naivasha, Sulunga, Kisale, Kabele, Kinkony)

SECTION 4. EMPIRICAL STRATEGY

The goal of this paper is to estimate the causal effect of lake water levels on local economic activity proxied by night-time lights. However, a simple OLS regression would fail due to endogeneity. Firstly, there is a concern of reverse causality, where economic activity may affect lake levels through water extraction. Secondly, omitted time-varying shocks, such as weather or demand-side commodity price booms. To address this, I exploit plausibly exogenous hydrological variation induced by rainfall in upstream catchments. Upstream rainfall affects lake levels hydrologically, but does not respond to contemporaneous local economic conditions. Further, I add fixed effects and local rainfall control to address the second channel of endogeneity.

Subsection 4.1. **Setup and notation.**

The dependent variable is NL_{it}: mean VIIRS-radiance in settled pixels within 10km ring.

The endogenous variables are:

- ZL_{it} : within-lake standardised levels

$$ZL_{it} = \frac{L_{it} - \bar{L}_i}{\sigma_{L_i}}.$$

- $Z\Delta L_{it}$: standardised first difference in raw lake levels

$$Z\Delta L_{it} = \frac{(L_{it} - L_{i,t-1}) - \overline{(L_{it} - L_{i,t-1})}}{\sigma_{\Delta L_i}}.$$

Instrumental variables are:
- $R^{up}_{i,t-1,12} \equiv \sum_{j=1}^{12} R^{up}_{i,t-j}$: 12-month cumulative rainfall in Lake i upstream catchment. Accumulation starts at lag 1 because lake level is measured at the start (or near) of month t, meaning rainfall in month t would only affect lake level $t+1$, not t.
- $R^{up}_{i,t-1},\ R^{up}_{i,t-2}$: Rainfall in lake i upstream catchment lag 1 and lag 2.

Controls are:
- $R^{ring}_{i,t-1,12} \equiv \sum_{j=1}^{12} R^{ring}_{i,t-j}$: 12-month cumulative rainfall in Lake i within the 10km ring.
- $R^{ring}_{i,t-1},\ R^{ring}_{i,t-2}$: rainfall in lake i within the 10km ring lag 1 and lag 2.

Fixed effects are:
- Lake fixed effects (γ_i) absorb all time-invariant differences across lakes. This includes baseline size, geography.
- Year fixed effects (λ_y) absorb shocks common to all lakes. This includes macro prices or conditions.
- Month of year fixed effects (μ_m) absorb seasonality shared across years, including wet and dry seasons. This, combined with local rainfall controls, ensures the results are not driven by seasonal trends.

All clustering of standard errors is done at the lake level, as outcomes and measurement errors are likely serially correlated within lakes.

Subsection 4.2. **Specification A - Stock Effects.**
This specification captures the effect of sustained changes in lake water level, showing a longer-run effect of hydrological conditions. The instrument is:

$$R^{up}_{i,t-1,12} \equiv \sum_{j=1}^{12} R^{up}_{i,t-j},$$

where the 12-month cumulative sum represents storage and persistence in lake hydrology.

First-stage equation:

$$ZL_{it} = \alpha^{(A1)} + \delta R^{up}_{i,t-1,12} + \theta R^{ring}_{i,t-1,12} + \gamma_i + \lambda_y + \mu_m + u_{it}.$$

Second-stage equation:

$$NL_{it} = \alpha^{(A2)} + \beta \widehat{ZL}_{it} + \theta R^{ring}_{i,t-1,12} + \gamma_i + \lambda_y + \mu_m + \varepsilon_{it}.$$

Here, β measures how one standard-deviation change in lake water level affects night-time lights, identified from persistent rainfall-driven differences in lake levels.

Subsection 4.3. **Specification B - Flow.**
This specification identifies lake levels using higher frequency inflow variation. Although the endogenous variable remains lake levels, the identification is driven by short-run upstream inflows. The instruments are $R^{up}_{i,t-1}$ and $R^{up}_{i,t-2}$.

First-stage equation:

$$ZL_{it} = \alpha^{(B1)} + \delta_1 R^{up}_{i,t-1} + \delta_2 R^{up}_{i,t-2} + \theta_1 R^{ring}_{i,t-1} + \theta_2 R^{ring}_{i,t-2} + \gamma_i + \lambda_y + \mu_m + u_{it}.$$

Second-stage equation:

$$NL_{it} = \alpha^{(B2)} + \beta \widehat{ZL}_{it} + \theta_1 R^{ring}_{i,t-1} + \theta_2 R^{ring}_{i,t-2} + \gamma_i + \lambda_y + \mu_m + \varepsilon_{it}.$$

Here, β represents the causal effect of a one within-lake standard-deviation increase in lake level on local night-time lights. The key is that this identifies a causal effect of transitory inflow-driven fluctuations in lake levels, whereas Specification-A captures the effect of more persistent, stock-like changes in levels.

Subsection 4.4. **Specification C - Change.**
This specification focuses on month-to-month shocks in lake levels. The endogenous regressor is the within-lake standardised change in lake level, $Z\Delta WL_{it}$. The instruments are $R^{up}_{i,t-1}$ and $R^{up}_{i,t-2}$.

First-stage equation:

$$Z\Delta L_{it} = \alpha^{(C1)} + \delta_1 R^{up}_{i,t-1} + \delta_2 R^{up}_{i,t-2} + \theta_1 R^{ring}_{i,t-1} + \theta_2 R^{ring}_{i,t-2} + \gamma_i + \lambda_y + \mu_m + u_{it}.$$

Second-stage equation:

$$NL_{it} = \alpha^{(C2)} + \beta \widehat{Z\Delta L}_{it} + \theta_1 R^{ring}_{i,t-1} + \theta_2 R^{ring}_{i,t-2} + \gamma_i + \lambda_y + \mu_m + \varepsilon_{it}.$$

Here, β represents the causal effect of a one-within-lake-standard-deviation month-to-month increase in lake levels on local night-time lights.

Subsection 4.5. **Instrument Validity and Inference.**
Instrument relevance is assessed using first-stage coefficients and partial F statistics.

Table 2: First-stage regressions

	Specification A: Stock	Specification B: Flow	Specification C: Change
Upstream rainfall (12m, lag 1)	0.001 81**		
	(0.000 59)		
Ring rainfall (12m, lag 1)	−0.000 55		
	(0.000 57)		
Upstream rainfall (lag 1)		−0.000 01	0.007 67***
		(0.001 85)	(0.001 35)
Upstream rainfall (lag 2)		0.004 24**	0.000 89
		(0.001 48)	(0.001 68)
Ring rainfall (lag 1)		0.000 72	−0.000 41
		(0.001 63)	(0.001 37)
Ring rainfall (lag 2)		0.000 23	−0.001 64
		(0.001 57)	(0.001 36)
Dependent variable	Lake level (within-lake z-score)	Lake level (within-lake z-score)	Δ lake level (within-lake z-score)
Num.Obs.	2024	2184	2184
Lake FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes
Clusters	lake_id	lake_id	lake_id
Partial F Statistic	9.60	10.73	214.61
Lakes	22	22	22

[1] + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

[2] Cluster-robust standard errors (lake) in parentheses. Excluded instruments are upstream rainfall measures (highlighted in bold); ring rainfall terms are included as controls. All specifications include lake, year, and month-of-year fixed effects.

Table2 shows positive first-stage coefficients on upstream rainfall and statistically significant joint tests of the excluded instruments. The F-statistic for spec C is much stronger than the conventional Staiger–Stock rule-of-thumb of 10 (Staiger and Stock 1997), but Spec A and B remain borderline but acceptable. I also complement my analysis with weak instrument procedures (Andrews, Stock, and Sun 2019)for robustness.

The exclusion restriction is supported by the spatial separation of the instrument and outcome, the local rainfall controls and fixed effects. Together, these

features substantially mitigate concerns that upstream rainfall affects local economic activity through other channels. Further, I restrict the sample to lakes with spatially separated outcome rings, reducing scope for spatial spillovers.

Section 5. Results

Subsection 5.1. Main IV estimates: Nighttime Lights.

Table 3: IV Estimates: Lake Levels and Nighttime Lights

	Specification A: Stock	Specification B: Flow	Specification C: Change
Lake level (within-lake z-score)	0.2822	0.9559***	
	(0.2566)	(0.1868)	
Δ lake level (within-lake z-score)			0.2489*
			(0.1193)
Ring rainfall (12m, lag 1)	−0.0006		
	(0.0004)		
Ring rainfall (lag 1)		−0.0014*	−0.0022**
		(0.0007)	(0.0008)
Ring rainfall (lag 2)		−0.0045***	−0.0003
		(0.0007)	(0.0005)
Dependent variable	Night-time lights (mean, 10km)	Night-time lights (mean, 10km)	Night-time lights (mean, 10km)
Independent variable	Lake level (within-lake z-score)	Lake level (within-lake z-score)	Δ lake level (within-lake z-score)
Num.Obs.	2024	2184	2184
Lake FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes
Clusters	lake_id	lake_id	lake_id
First-stage partial F (excluded IVs)	9.60	10.73	214.61
Lakes	22	22	22

[1] + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

[2] Specification A instruments standardised lake level with upstream rainfall accumulated over the previous 12 months (lag 1), controlling for ring rainfall (12m, lag 1). Specification B instruments standardised lake level with upstream rainfall lags 1 and 2, controlling for ring rainfall lags 1 and 2. Specification C instruments the standardised month-to-month change in lake levels with upstream rainfall lags 1 and 2 (ring rainfall lags 1 and 2 as controls). Bolding indicates endogenous regressors. Standard errors are clustered by lake.

Specification-A gives us a positive but statistically insignificant coefficient on standardised lake levels. The estimate of 0.28 suggests that higher long-run lake water stock is associated with higher local nighttime lights but estimated imprecisely.

Specification-B shows a large and significant positive effect of lake levels on night-time lights. A one within-lake-standard-deviation increase in lake levels leads to an increase of approximately 0.96 units in mean nighttime lights (0.30 standard-deviation of monthly nightlights (SD = 3.259)), suggesting that local economic activity is strongly responsive to higher frequency changes in lake levels, adding this as a contribution to the literature.

Specification-C focuses on month-to-month changes in lake level and provides strong evidence. The estimated coefficient is positive and significant at the 5%, implying a one within-lake-standard-deviation increase in the monthly lake level change raises local night-time lights by 0.25 units.

Overall, these results suggest that local economic activity responds strongly to short-run, inflow-driven changes in lake level, which has been previously understudied. This pattern is consistent with mechanisms that operate at high frequency, including irrigation or transport.

Subsection 5.2. **Robustness Checks.**

5.2.1. *Weak Instrument Procedures.*
In-line with weak-instrument literature, I report the Anderson-Rubin (AR) 95% Confidence Intervals and the CLR confidence intervals, which show what effect sizes remain consistent with the data and whether no effect can be ruled out (Staiger and Stock 1997; Andrews, Stock, and Sun 2019).

Table 4. Weak-Instrument–Robust Inference (Specification A)

Test	**Test Statistic**	**95% Confidence Interval**
Anderson–Rubin (F)	$F = 3.716\ (1, 2022)$	$[-0.005,\ 0.635]$
Conditional Likelihood Ratio (CLR)	3.716	$[-0.005,\ 0.635]$

Notes: The table reports weak-instrument–robust inference for Specification A. Both the Anderson-Rubin and Conditional Likelihood Ratio tests evaluate the null hypothesis $H_0 : \beta = 0$. Reported confidence intervals are robust to weak instruments. p-values for both tests equal 0.054.

For Specification-A, the confidence intervals include 0, although the lower bound lies close to 0. The AR test rejects the null at the 10%, suggesting a positive but imprecise estimated effect.

Table 5. Weak-Instrument–Robust Inference (Specification B)

Test	**Test Statistic**	**95% Confidence Interval**
Anderson–Rubin (F)	$F = 26.57\ (2, 2181)$	$[0.603,\ 1.613]$
Conditional Likelihood Ratio (CLR)	52.51	$[0.650,\ 1.482]$

Notes: The table reports weak-instrument–robust inference for Specification B. Both tests evaluate the null hypothesis $H_0 : \beta = 0$. The Anderson–Rubin statistic follows an $F(2, 2181)$ distribution under the null, whilst the Conditional Likelihood Ratio test uses a normal approximation. Confidence intervals are robust to weak instruments.

In contrast, Spec B results are robust with confidence intervals excluding zero by a wide margin. The AR test strongly rejects the null of no effect, supporting the positive effect that short-run lake levels have on nighttime lights.

5.2.2. *20km Outcome ring.*
This tests whether estimated effects are sensitive to spatial space by re-estimating the main specification using a 20km lake ring.

IV Estimates: Lake Levels and Nightlights (VIIRS, Mean within 20km Ring)

Table 6:

	Specification A: Stock	Specification B: Flow	Specification C: Change
Lake level (within-lake z-score)	0.1251	0.8628***	
	(0.2377)	(0.1622)	
Δ Lake level (within-lake z-score)			0.2026*
			(0.0934)
Ring rainfall (12m, lag 1)	0.0000		
	(0.0004)		
Ring rainfall (lag 1)		−0.0013+	−0.0019**
		(0.0007)	(0.0007)
Ring rainfall (lag 2)		−0.0041***	−0.0003
		(0.0006)	(0.0004)
Independent variable	Lake level (z-score)	Lake level (z-score)	Δ Lake level (z-score)
Dependent variable	VIIRS nightlights (mean, 20km ring)	VIIRS nightlights (mean, 20km ring)	VIIRS nightlights (mean, 20km ring)
Num. Obs.	2024	2184	2184
Lake FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes
Clusters	lake_id	lake_id	lake_id
First-stage partial F (excluded IVs)	9.60	10.73	214.61
Lakes	22	22	22

[1] + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
[2] Specification A instruments within-lake standardised lake level using upstream rainfall accumulated over 12 months (lagged 1 month), controlling for the analogous ring rainfall measure. Specification B instruments levels using upstream rainfall lags 1 and 2 months (ring lags included as controls). Specification C instruments within-lake standardised month-to-month changes in lake levels using upstream rainfall lags 1 and 2 months (ring lags included as controls). Standard errors are clustered by lake. Bolding indicates independent variable.

Table6 shows a positive but insignificant coefficient with Specification-A. Specification-B provides a significant and positive coefficient of 0.86, indicating a weaker effect than with 10km. Specification-C also provides a positive and significant coefficient, proving effects still persist at a broader scale but have stronger responses by proximity to the lake.

5.2.3. *Vegetation Mechanism.*
To test a potential mechanism, I change the outcome to satellite-derived Normalised Difference Vegetation Index (NDVI).

IV Estimates: Lake Levels and NDVI

Table 7:

	Specification A: Stock	Specification B: Flow	Specification C: Change
Lake level (within-lake z-score)	0.0490**	0.0924**	
	(0.0174)	(0.0326)	
Change in lake level (within-lake z-score)			0.0620***
			(0.0116)
Ring rainfall (12m, lag 1)	0.0000		
	(0.0000)		
Ring rainfall (lag 1)		0.0003**	0.0000
		(0.0001)	(0.0001)
Ring rainfall (lag 2)		−0.0002	0.0002**
		(0.0001)	(0.0001)
Independent variable	Lake level (z-score)	Lake level (z-score)	Δ Lake level (z-score)
Dependent variable	NDVI (mean, 10km buffer)	NDVI (mean, 10km buffer)	NDVI (mean, 10km buffer)
Num. Obs.	2024	2184	2184
Lake FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes
Clusters	lake_id	lake_id	lake_id
First-stage partial F (excluded IVs)	9.60	10.73	214.61
Lakes	22	22	22

[1] + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
[2] Specification A instruments within-lake standardised lake level using upstream rainfall accumulated over 12 months (lagged 1 month), controlling for the analogous ring rainfall measure. Specification B instruments levels using upstream rainfall lags 1 and 2 months (ring lags included as controls). Specification C instruments within-lake standardised month-to-month changes in lake levels using upstream rainfall lags 1 and 2 months (ring lags included as controls). Standard errors are clustered by lake.

This indicates that higher lake levels lead to increased local vegetation, evidencing a supply-side mechanism. Across all specifications, increases in lake levels are associated with statistically significant rises in NDVI. Most importantly, Specification-A shows a positive and precisely estimated effect consistent with vegetation responding strongly to sustained changes in surface water availability. These findings suggest that an increase in lake levels has both longer-term and short-term positive effects on agriculture.

Subsection 5.3. **Heterogeneity.**

5.3.1. *Lake Size.*
To explore the effects of lake levels on the physical scale of the lake itself, I split the sample into large and small by median lake surface area.

IV Estimates: Lake Levels and Nighttime Lights — Heterogeneity by Lake Size

	Spec A: Small	Spec A: Large	Spec B: Small	Spec B: Large	Spec C: Small	Spec C: Large
(Intercept)	−3.506	0.783*	−0.904	0.672**	−0.272***	0.282
	(6.286)	(0.369)	(0.668)	(0.246)	(0.058)	(0.145)
Lake level (z-score)	−1.238	0.475*	−0.359	0.940***		
	(2.362)	(0.192)	(0.367)	(0.199)		
Δ Lake level (z-score)					−0.217	0.339**
					(0.173)	(0.109)
Ring rainfall (12-month, lag 1)	0.002	−0.001*				
	(0.005)	(0.000)				
Ring rainfall (lag 1)			0.001	−0.001	0.002	−0.003***
			(0.001)	(0.001)	(0.001)	(0.001)
Ring rainfall (lag 2)			0.002	−0.004***	−0.001	−0.001
			(0.002)	(0.001)	(0.000)	(0.001)
Independent variable	Lake level	Lake level	Lake level	Lake level	Δ Lake level	Δ Lake level
Dependent variable	Nighttime lights	Nighttime lights	Nighttime lights	Nighttime lights	Nighttime lights	Nighttime lights
Num. Obs.	869	1155	969	1215	969	1215
First-stage F-statistic	1.90	28.66	1.07	22.57	9.46	72.96
Lakes	11	11	11	11	11	11

[1] * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

[2] Cluster-robust standard errors (lake) in parentheses. All regressions include lake, year, and month-of-year fixed effects. Specification A instruments within-lake standardised lake level using upstream rainfall accumulated over 12 months (lag 1), controlling for ring rainfall (12-month, lag 1). Specification B instruments levels using upstream rainfall lags 1–2 months (ring rainfall lags included as controls). Specification C instruments month-to-month changes in lake levels using the same rainfall lags. Sample split by median lake surface area (km^2).

The results indicate that estimated effects are driven by larger lakes showing they are more important to local economic activity. In all specifications, lake level changes have no detectable effect for smaller lakes but show a large and precise positive effect for larger lakes. This coincides with stronger F-statistics in larger lakes, due to larger upstream rain-catchments.

5.3.2. *Seasonality.*
This test focuses on lake level effects on local economic activity during a dry season, when communities may be more constrained. To define a dry month, I compute the average local rainfall by month-year and classify dry months as below-median.

	Specification A (Stock)	Specification B (Flow)	Specification C (Change)
Lake level (z-score)	−0.517	1.183*	
	(0.522)	(0.532)	
Lake level (z-score) × Dry month	1.777	−0.993	
	(1.202)	(1.302)	
Δ Lake level (z-score)			0.975*
			(0.427)
Δ Lake level (z-score) × Dry month			−1.883**
			(0.678)
Ring rainfall (12-month, lag 1)	−0.001		
	(0.000)		
Ring rainfall (12-month, lag 1) × Dry	0.000		
	(0.000)		
Ring rainfall (lag 1)		−0.001	−0.004*
		(0.001)	(0.002)
Ring rainfall (lag 2)		−0.004**	0.001
		(0.002)	(0.001)
Ring rainfall (lag 1) × Dry		0.000	0.008***
		(0.001)	(0.002)
Ring rainfall (lag 2) × Dry		0.000	−0.005**
		(0.003)	(0.002)
Independent variable	Lake level (z-score)	Lake level (z-score)	Δ Lake level (z-score)
Dependent variable	Nighttime lights	Nighttime lights	Nighttime lights
Num. Obs.	2024	2184	2184
Lakes	22	22	22

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
Cluster-robust standard errors (lake) in parentheses. All specifications include lake, year, and month-of-year fixed effects. Dry months are defined lake-by-lake as months-of-year with below-median average ring rainfall. Specification A instruments lake levels using upstream rainfall accumulated over 12 months (lag 1), controlling for ring rainfall (12-month, lag 1). Specification B instruments lake levels using upstream rainfall lags 1–2 months. Specification C instruments month-to-month changes using the same lags. Interaction terms are instrumented using the corresponding upstream rainfall terms interacted with the dry-month indicator; ring rainfall terms and their interactions are included as controls.

Across specifications A & B, the estimated interaction is negative but insignificant. In Specification-C, the month-to-month increases in lake levels lead to higher economic activity, but this effect is significantly lower in dry months. This

is likely because of anticipation where water inflows are less productive in dry months due to constraints on agricultural cycles, infrastructure or demand.

5.3.3. *Baseline Nightlights Radiance.*
This test focuses on the heterogeneity of effects on areas that have a higher baseline level of nightlight radiance. Lakes are classified as high or low based on their average night-time lights in 2012 (Start of sample).

	Specification A: Stock	Specification B: Flow	Specification C: Change
Lake level (z-score) [Low baseline NL]	**0.4253**	**0.5956*****	
	(0.5389)	(0.1783)	
Additional effect in High baseline NL	**-0.1555**	**0.3252****	**-0.0372**
	(0.5386)	(0.1506)	(0.0456)
Δ Lake level (z-score) [Low baseline NL]			**0.2453****
			(0.0985)
First-stage F (levels)	6.180	6.160	55.51
First-stage F (interaction)	6.890	20.57	291.1
Baseline NL split year	2012	2012	2012
Lake FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes
Independent variable	Lake level (z-score)	Lake level (z-score)	Δ Lake level (z-score)
Dependent variable	Nighttime lights (mean, 10km)	Nighttime lights (mean, 10km)	Nighttime lights (mean, 10km)
Observations	2,024	2,184	2,184

Clustered (lake_id) standard errors in parentheses.
Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The results indicate that the economic response to lake levels is substantially stronger in areas with higher baseline economic activity, but with low F-statistics for lower areas. This could be because higher baseline areas are better able to respond to higher lake levels as they have more capital to take advantage.

Subsection 5.4. Alternative Outcome: Conflict.

To examine whether changes in lake levels affect local conflict, I change the outcome to $log(1 + events_{it})$ within the 10km ring.

IV Estimates: Lake Levels and Conflict (ACLED, 10km ring)

Table 8:

	Spec A: Stock	Spec B: Flow	Spec C: Change
Lake level (within-lake z-score)	0.1929*	−0.0566	
	(0.0846)	(0.0304)	
Change in lake level (within-lake z-score)			−0.0324***
			(0.0092)
Ring rainfall (12m, lag 1)	−0.0003*		
	(0.0001)		
Ring rainfall (lag 1)		0.0001	0.0003**
		(0.0001)	(0.0001)
Ring rainfall (lag 2)		0.0002*	0.0000
		(0.0001)	(0.0001)
Independent variable	Lake level (z-score)	Lake level (z-score)	Δ Lake level (z-score)
Dependent variable	ACLED: log(1 + events) (10km)	ACLED: log(1 + events) (10km)	ACLED: log(1 + events) (10km)
Num. Obs.	2024	2184	2184
Lake FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Month-of-year FE	Yes	Yes	Yes
Clusters	lake_id	lake_id	lake_id
First-stage partial F Statistic	9.60	10.73	214.61
Lakes	22	22	22

[1] + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

[2] Spec A instruments within-lake standardised lake level using upstream rainfall accumulated over 12 months (lagged 1 month), controlling for the analogous ring rainfall measure. Spec B instruments levels using upstream rainfall lags 1 and 2 months (ring lags included as controls). Spec C instruments within-lake standardised month-to-month changes in lake levels using upstream rainfall lags 1 and 2 months (ring lags included as controls). Standard errors are clustered by lake.

The results indicate that short-term increases in lake level lead to fewer conflict events, as Specification-C shows. This highlights short-term increases in lake level might lead to alleviating pressure on livelihoods, causing fewer conflicts. However, a positive and marginally significant coefficient with Specification-A reflects that longer-term increases in lake level likely lead to more urbanisation, meaning more conflicts.

Section 6. Conclusion

This paper provides causal, monthly evidence on lake water levels and economic activity across multiple regions using an IV strategy. It adds to the literature through moving beyond single-lake or long-run historical analyses, showing that short-run inflow-driven variation matters significantly to local economic activity proxied by night-time lights. The paper shows these effects are economically meaningful and robustly supported by mechanism and conflict evidence.

Nighttime lights imperfectly measure economic activity (noise may attenuate coefficients toward zero) and the rainfall IV identifies a local average treatment effect relevant to only inflow-driven variation. Instrument strength is borderline in some specifications and heterogeneity splits, which motivates weak-instrument-robust-inference but limits precision. External validity-wise, the sample is restricted to undammed lakes in Africa and South America, so the results may not generalise to regulated reservoirs or highly urbanised regions. Whilst the

monthly panel captures short-run responses, it doesn't directly analyse long-run adaptation effects.

Future work could link lake-level variation to outcomes like wages, employment, or household welfare to capture richer mechanisms. A key extension is to analyse non-linear responses and threshold effects, testing if effects change with lake-levels crossing critical values. Research could also explore adaptive responses to lake levels, seeing if firms adjust production or investment decisions to dampen exposure. Finally, combining this paper with climate projections could help identify regions most vulnerable to surface-water loss to inform climate adaptation and water management policy.

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